FINAL PROJECT DEMONSTRATION

Design and Simulation of Emergency Alarm System

MADHU PRIYA
2620080072

VENKATA RAMANA
2620080054

JEYANTH VIGHNESH
2620080063

JUHITHA
2620080169

Section: [User Section]

Roll Numbers: 0072, 0054, 0063, 0169

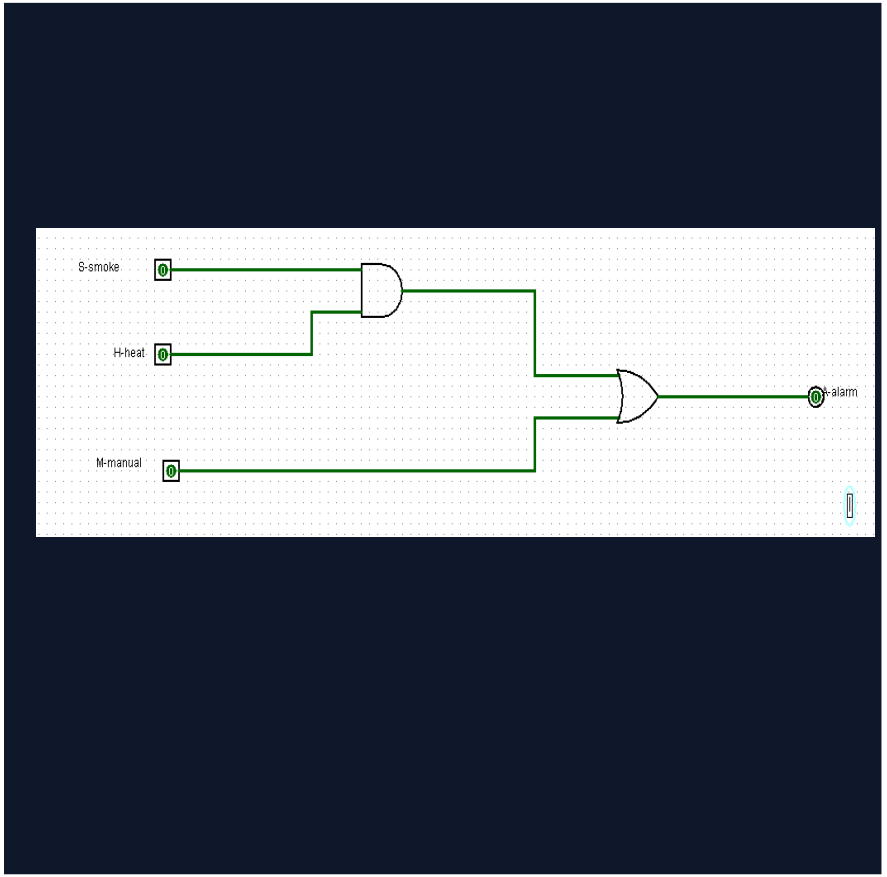
INTRODUCTION & OBJECTIVE

Introduction

Emergency alarm systems are critical safety components in industrial, residential, and commercial environments. They utilize digital logic to monitor sensors and trigger immediate alerts during hazardous conditions.

Project Objectives

- Design a robust digital logic-based alarm trigger.
- Simulate sensor-driven emergency scenarios.
- Ensure high-reliability and low-latency response.
- Implement cost-effective safety monitoring using standard logic gates.



THEORY & LOGIC CIRCUIT

Basic Concept

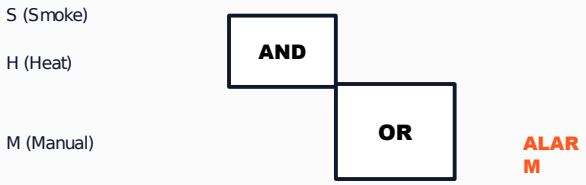
The system uses three primary inputs representing sensors: Smoke (S), Heat (H), and Manual Trigger (M). The alarm activates if there is both smoke and heat, OR if the manual button is pressed.

Boolean Expression

$$Alarm = (S \text{ AND } H) \text{ OR } M$$

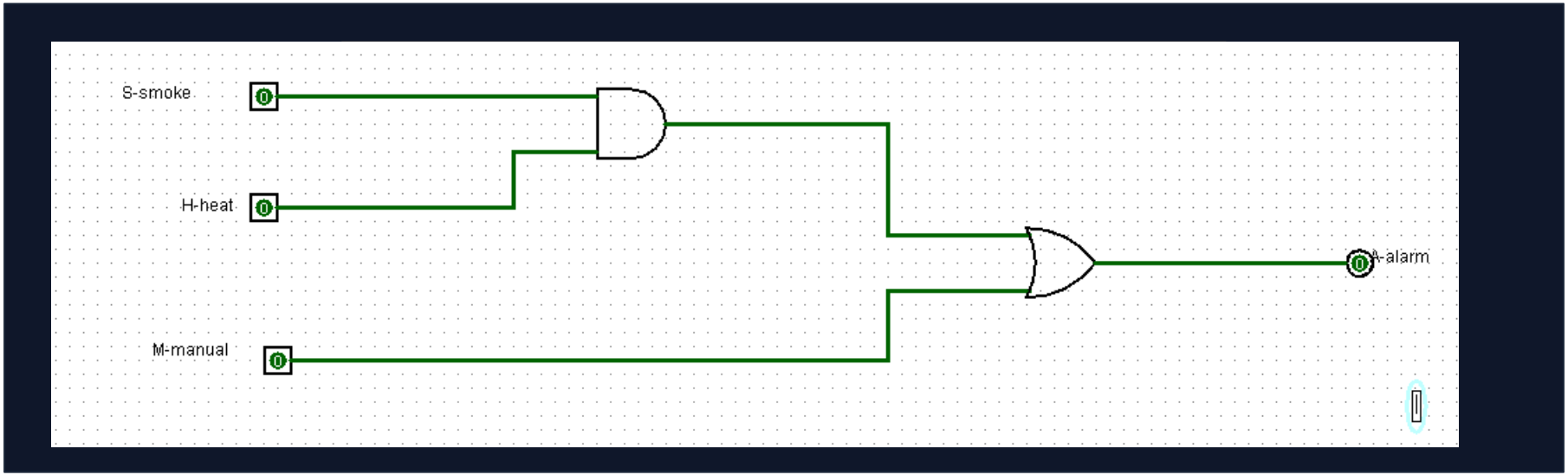
Logic Gates Used

- **AND Gate:** For sensor coincidence.
- **OR Gate:** For combining triggers.
- **NOT Gate:** (Optional) For inverted status.



Functional Schematic Representation

CIRCUIT DESIGN IN LOGISIM



Input Pins

Discrete digital pins represent sensors. High state (1) denotes detection, Low (0) denotes idle.

Signal Flow

The AND gate validates simultaneous sensor trips before passing the signal to the final OR gate.

Output Buzzer

A single output pin drives the alarm indicator, visualized as a red LED in the simulation.

SIMULATION RESULTS

Truth Table

S	H	M	Alarm
0	0	0	0
0	1	0	0
1	0	0	0
1	1	0	1
0	0	1	1
0	1	1	1
1	0	1	1
1	1	1	1

Case 1: Sensor Detection

When S=1 and H=1, the AND gate output becomes 1, triggering the alarm through the OR gate. This prevents false alarms from heat or smoke alone.

Case 2: Manual Trigger

Setting M=1 bypasses sensor logic. The OR gate ensures the alarm activates immediately regardless of sensor states.

Simulation Stability

The circuit maintains stable outputs across all 8 input combinations without race conditions or signal glitching.

CONCLUSION & APPLICATIONS

Project Summary

The Emergency Alarm System successfully demonstrates the application of basic digital logic gates in solving real-world safety challenges. Through Logisim simulation, we verified that the system operates reliably under both sensor-driven and manual trigger conditions.

Industrial Safety

Residential Alarms

Data Centers

Healthcare Units

Future Improvements

Integration with IoT for remote monitoring.

Addition of Gas and Water Leakage sensors.

Implementation of Latch circuits to keep alarm active.

Voice-based alert system integration.

Power backup simulation using UPS models.

THANK YOU